LEVEL UP ACADEMY

Class 10 | GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 12 Test (GSEB pattern): Magnetic Effects of Electric Current

Chapter / Topic: Magnetic Effects of Electric Current

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SECTION A

Q1. Which of the following correctly describes the magnetic field near a long, straight, current-carrying wire? **[1]**

Answer: (c) Concentric circles centred on the wire

Q2. Outside a bar magnet, magnetic field lines go: **[1]**

Answer: (a) from the north pole to the south pole

Q3. The magnetic field inside a long, straight, current-carrying solenoid: **[1]**

Answer: (d) is the same at all points (uniform)

Q4. A current-carrying conductor placed parallel to a magnetic field experiences: **[1]**

Answer: (b) no force

Q5. A positively charged particle (alpha particle) moving towards the west is deflected towards the north by a magnetic field. The direction of the… **[1]**

Answer: (c) upward (by Fleming's left-hand rule)

Q6. Which instrument detects the presence of a current in a circuit? **[1]**

Answer: (a) Galvanometer

Q7. Magnetic field lines: **[1]**

Answer: (b) never intersect each other

Q8. Which of the following increases the strength of an electromagnet? **[1]**

Answer: (d) Using a soft iron core

Q9. Which device protects a household circuit from short-circuiting and overloading? **[1]**

Answer: (a) Fuse (or MCB)

SECTION B

Q10. State Fleming's left-hand rule. **[2]**

Answer: Stretch the thumb, forefinger and middle finger of the left hand so that they are mutually perpendicular. If the forefinger points in the direction of the magnetic field and the middle finger in the direction of the current, then the thumb points in the direction of the force (motion) on the conductor.

- Three fingers mutually perpendicular: 1
- Correct assignment of field, current, force: 1

Q11. Describe the magnetic field lines of a current-carrying circular loop. Why is the field strong at the centre? **[2]**

Answer: Near the wire, the field lines are concentric circles, which grow larger farther from the wire. At the centre of the loop they appear as nearly straight lines. Every part of the loop produces a field in the same direction at the centre, so the fields add up and make it strong there.

- Pattern: 1
- Fields add at centre: 1

Q12. What is the function of the earth wire? Why should metallic appliances be earthed? **[2]**

Answer: The earth wire is a safety measure that gives a low-resistance path for current. If current leaks to the metal body of an appliance, it flows to the earth, keeping the body at earth potential. This prevents a severe electric shock to the user.

- Low-resistance path to earth: 1
- Prevents shock from leakage: 1

SECTION C

Q13. Answer the following about a solenoid: **[3]**

(i) A coil of many circular turns of insulated copper wire, wrapped closely in the shape of a cylinder. [1]

(ii) The field patterns are similar. One end of the solenoid behaves as a north pole and the other as a south pole. [1]

(iii) Place a soft iron core inside the solenoid and pass a current through it. The core becomes strongly magnetised. [1]

Q14. A circular loop of wire lies flat on a table, and current flows through it clockwise (as seen from above). **[3]**

(i) If you hold a current-carrying conductor in your right hand with the thumb along the current, your curled fingers show the direction of the magnetic field lines. [1]

(ii) Perpendicular to the table, pointing into it (downwards). [1]

(iii) Perpendicular to the table, pointing out of it (upwards). [1]

SECTION D

Q15. Rohan's house has two separate circuits: one rated 15 A for high-power appliances like the geyser and AC, and one rated 5 A for fans and bulbs. One… **[4]**

(i) High-power appliances draw a large current, so a separate high-rating circuit prevents overloading of the lighting circuit. [1]

(ii) Overloading. The total current drawn was more than the rated value of the circuit. [1]

(iii) $I = P/V = 2000/220 \approx 9.1$ A. This is more than 5 A, so it should not be used on the 5 A circuit. Use the 15 A circuit. [2]

- $I \approx 9.1$ A: 1
- Not safe on 5 A, with reason: 1